AYMANE AIT DADS

Time Series Machine Learning Intern | Predictive Modeling · Feature Engineering · ML Pipeline Development

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, $106K+ prize pool) by designing and executing an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer model with BF16 mixed precision and 8,192-token context windows. Reduced manual network analysis time by 60% at Orange Maroc by building automated Random Forest and K-Means pipelines on 100K+ fiber-optic records, directly informing network optimization decisions. Proficient in Python, PyTorch, Scikit-learn, Pandas, and feature engineering workflows - skills directly applicable to time series forecasting, anomaly detection, and industrial machine learning at Schneider Electric. Eager to apply structured experimentation and production-oriented ML pipeline design to connected energy systems and operational automation challenges.

CORE COMPETENCIES

Time Series Analysis | Predictive Modeling | Feature Engineering | Anomaly Detection | Machine Learning Pipeline | Python & Scikit-learn | Data Visualization | Model Evaluation

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built Random Forest classification and K-Means clustering models on **100K+ fiber-optic network records** - including upload speed, latency, and jitter - to classify network quality across thousands of nodes.
- Developed an end-to-end clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation workflows applied to large-scale network time series data.
- Delivered stakeholder presentations translating model outputs into actionable maintenance recommendations, using **Power BI** dashboards for operational data visualization.

PROJECTS

Anomalous Sound Detection - Industrial Equipment
EURECOM · 2025 · AUC 0.80 Predictive Maintenance

- Designed an unsupervised fault detection system for slide-rail industrial machinery using a Transformer autoencoder on **Mel spectrograms with SpecAugment**, achieving AUC 0.80 with no labeled anomaly data.
- Built a complete **anomaly detection pipeline** covering spectrogram extraction, augmentation, autoencoder training, and reconstruction-error thresholding for predictive maintenance in industrial settings.

PyTorch · Transformer Autoencoder · Mel Spectrograms · SpecAugment · Scikit-learn

NVIDIA Nemotron Model Reasoning Challenge
Kaggle · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context, grad_accum=32).
- Ran **controlled ablation experiments** across learning rate schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM
Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya
2021 - 2023

SKILLS

ML Frameworks & Engineering: Python | PyTorch | Scikit-learn | TensorFlow | Pandas | NumPy | Feature Engineering

Machine Learning Methods: Time Series Analysis | Anomaly Detection | Random Forest | Clustering | Ensemble Methods | LoRA SFT | Predictive Modeling

Data & Visualization: Power BI | Matplotlib | Seaborn | EDA | SQL | Git | Linux